EX NO: 7

DATE: 20/8/26

DEVELOP A SIMPLE CALCULATOR USING XMLRPC

AIM:

To develop a Simple Calculator using XMLRPC (XML Remote Procedural Call)

INTRODUCTION TO XML-RPC:

XML-RPC (Remote Procedural Call) is a protocol that allows a client application to execute a function or procedure on a remote Server over a network. It uses XML to encode the request and response messages and commonly works over HTTP for Communication. XML-RPC follows a Simple Client-Server architecture. Where the client Sends a request containing the method name and parameters and the Server processes the request and sends back the result. It can be used to build distributed applications such as a remote calculator, when arithmetic operations are performed by the Server based on request from the client.

PROCEDURE:

1. Create an XML-RPC Server that provides methods for Basic addition, Subtraction, multiplication and division
2. Start the Server on a Specified host and port
3. Create an XML-RPC client to connect the Server.
4. Get the required numbers and operation from the User.

5. Send the request from the client to the Server.
6. The Server performs the Calculation and returns the result.
7. Display the result on the Client Side
8. Test the Calculator with different inputs.

SERVER SIDE PROGRAM:

```
from xmlrpc.server import SimpleXMLRPCServer

def add(x,y): return x+y
def subtract(x,y): return x-y
def multiply(x,y): return x*y
def divide(x,y): return x/y if y!=0 else "Error: Div by 0"

server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Calculator Server is running on port 8000...")
server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")
server.serve_forever()
```

OUTPUT

```
xmlrpc Calculator client
  1. Add
  2. Subtract
  3. Multiply
  4. Divide
  5. Exit
Choose an operation (1-5)
  1
Enter first number : 20
Enter Second number : 30
Result : 50
  2
Enter first number : 5
Enter Second number : 2
Result : 3
  3
Enter first number : 5
Enter Second number : 4
Result : 20
  4
Enter first number : 10
Enter Second number : 2
Result : 5
  5
Exiting
```

CLIENT SIDE PROGRAM:

```
import xmlrpc.client
server = xmlrpc.client.ServerProxy("http://localhost:8000")
while True:
print ("\n--XML-RPC Calculator Client--")
print ("1. Add")
print ("2. Subtract")
print ("3. Multiply")
print ("4. Divide")
print ("5. Exit")
choice = input("choose an operation(1-5): ")
if choice == '5':
    print("Exiting Calculator. Goodbye")
    break
if choice not in ['1','2','3','4']:
    print("Invalid choice! Please select 1,2,3,4 or 5") continue

try:
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
except ValueError:
    print("Error: Please enter valid numbers!") continue

if choice == '1':
    print("Result:", server.add(num1, num2))
elif choice == '2':
    print("Result", server.subtract(num1, num2))
elif choice == '3':
    print("Result:", server.multiply(num1, num2))
elif choice == '4':
    print("Result:", server.divide(num1, num2))
```

RESULT:

Thus the Simple Calculator is implemented using XML RPC